

## Part B Question 2

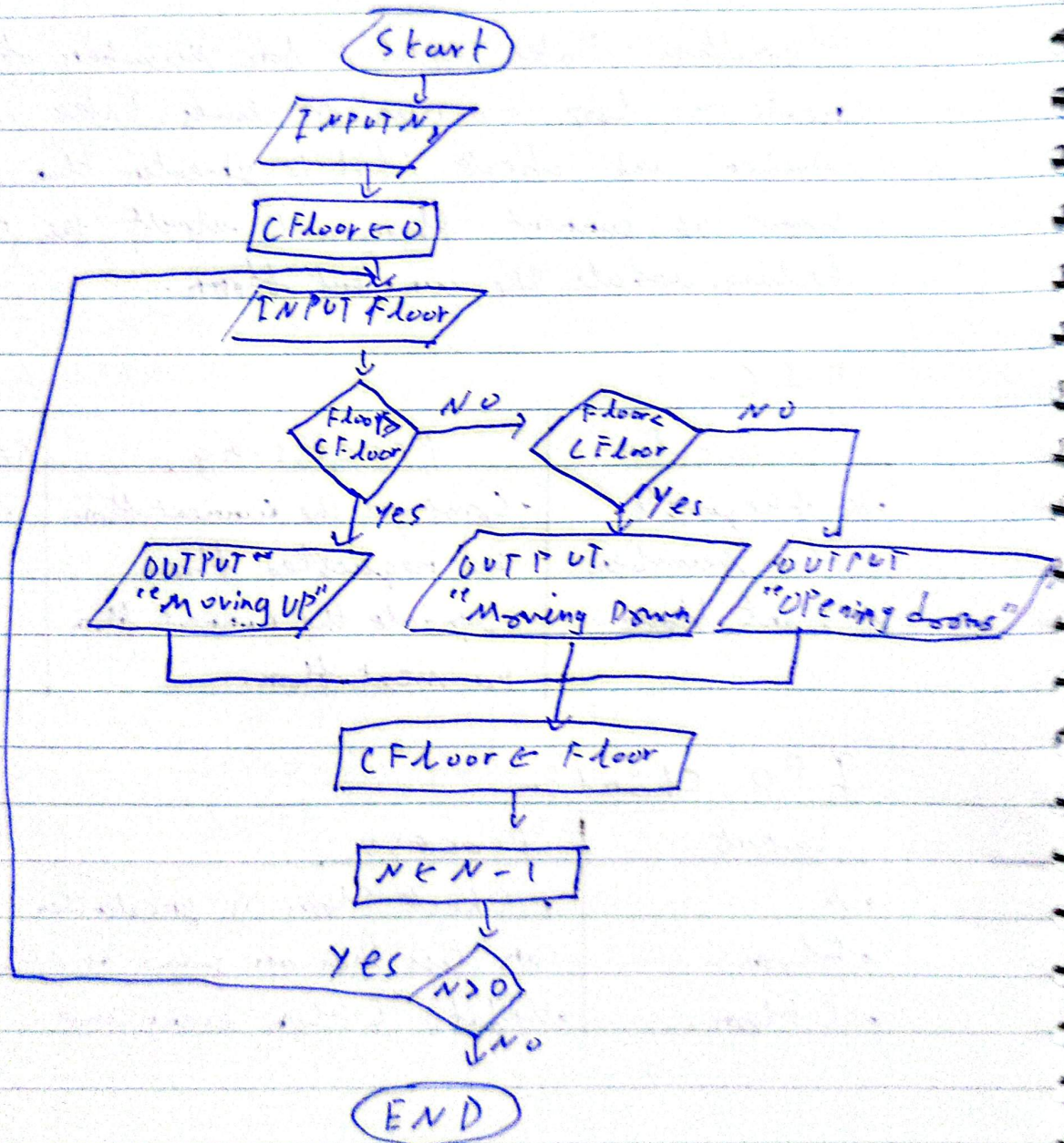
Algorithm:- Take input  $N$  for Number of requests. Create a loop to run  $N$  times. Take input floor number and check if it is greater than or less than or same as current floor and output the result after checking. Update the current floor.

PAC:-

| Given Data                                                                                                   | Processing                                                                                                                                        | Required Result                                                     |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>No. of Requests</li> <li>Floor number</li> <li>Start floor</li> </ul> | <ul style="list-style-type: none"> <li>Compare the current floor and requested floor.</li> <li>Update the current floor to next floor.</li> </ul> | <ul style="list-style-type: none"> <li>Movement of lifts</li> </ul> |

IPO Chart:-

| Input                                                                                            | Process                                                                                                                                          | OUTPUT                                                              |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li><math>N</math></li> <li>Floor</li> <li>C Floor</li> </ul> | <ul style="list-style-type: none"> <li>Check Floor is greater than or less than or same as C Floor</li> <li>update C Floor every time</li> </ul> | <ul style="list-style-type: none"> <li>Movement of lifts</li> </ul> |

Flow Chart :-

Pseudocode :-

INPUT  $N$

$CFloor \leftarrow 0$

WHILE  $N > 0$  DO,

INPUT ~~Floor~~ Floor

IF Floor  $>$  CFloor Then

OUTPUT "moving UP"

ELSE IF Floor  $<$  CFloor Then

OUTPUT "moving Down"

ELSE

OUTPUT "Doors opening"

ENDIF

$CFloor \leftarrow Floor$

$N \leftarrow N - 1$

END WHILE